



# Armada Kalamunda Group

## Surgical Prophylaxis Guideline

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### 1. Purpose

Administration of antibiotics before surgery has shown to reduce risk of post-operative infections.

This Guideline outlines the current Australian recommendations detailed in the Electronic Therapeutic Guidelines (eTG) and its application in the Armada Kalamunda Group (AKG).

### 2. Applicability

This Guideline applies to Medical and Surgical procedures as outlined in Appendix 1 performed at Armada Kalamunda Group.

### 3. Guidelines

- Surgical antibiotic prophylaxis **MUST** be administered **before** surgical incision.
  - **Short-action antibiotics (e.g. cefaxolin):** dose must be administered no more than 60 minutes before incision.
  - **Antibiotics that are not short-acting (e.g. metronidazole, clindamycin):** dose must be administered no more than 120 minutes before surgical prophylaxis.
  - **Vancomycin:** requires a slow infusion. Start vancomycin infusion within the 15 to 120 minutes before surgical incision. Surgical incision can occur before the infusion is completed.
- Postoperative (intravenous or oral) antibiotic therapy should be avoided and does NOT provide better patients outcomes than a single pre-operative dose for majority of procedures.
- Surgical antibiotic prophylaxis should **NOT** generally extend post-operatively beyond 24
  - Postoperative doses of IV antibiotics of up to 24 hours may only be considered for a small minority of procedures (identified in relevant guidelines, e.g. total knee arthroplasty).
  - Do NOT extend the duration of antibiotic prophylaxis because the patient has a urinary or intravascular catheter, or surgical drain, *in situ*.

#### 3.1 Surgical antibiotic prophylaxis guidelines – contained in appendix 1 attached

- Abdominal Surgery
- Appendectomy procedures
- Biliary Surgery, including laparoscopic surgery
- Breast Surgery
- Colorectal Surgery
- Ear Nose and Throat Surgery

- Endoscopic Retrograde Cholangiopancreatography
- Endoscopic Ultrasound
- Endoscopic Urological Procedures
- Head and Neck surgery
- Hernia Repair
- Implantable Cardiac Device Insertion
- Neurosurgery Procedures
- Obstetric Surgery
- Open or laparoscopic urological procedures
- Ophthalmic Surgery
- Oral Maxillofacial procedures
- Orthopaedic Surgery
- Skin and Soft Tissue Procedures
- Small Intestinal Surgery
- Spinal Surgery
- Transperineal prostate biopsy
- Transrectal prostate biopsy
- Transurethral resection of the prostate
- Vascular Surgery

**A detailed allergy history should always be obtained.**

**Details should include the type of antibiotic, the clinical details of the reaction and when the reaction occurred.**

#### 4. Roles and responsibilities

It is the responsibility of both anaesthetist and surgeon to prescribe the appropriate antibiotic and ensure its correct administration before surgical incision as per the Surgical Antibiotic Prophylaxis Guidelines (Appendix 1)

| Role                          | Responsibility                                                                                                                                                                     |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AMS Pharmacist                | To ensure adequate stock levels of the requisite antibiotics; if there is a shortage, a discussion between ID and Pharmacy to ensure an adequate supply of an alternate antibiotic |
| Infectious Disease Consultant | To facilitate ongoing monitoring through regular stewardship rounds to pick up issues when surgical prophylaxis is continued > 24 hours                                            |

#### 5. Legislative context

The following documents are required to give effect to this policy:

- *Medicines and Poisons Act 2014*
- *Medicines and Poisons Regulations 2016*

- Department of Health Western Australia (HDWA) [High Risk Medication Policy MP 0131/20](#)

## 6. Supporting information

The following documents support the implementation of this policy:

- HDWA [Statewide Medicines Formulary Policy MP 0077/18](#)
- [Medication Management Policy AKG-POL-0253-13](#)
- [Medication Management Procedure AKG-PRO-0255-18](#)
- HDWA [Guiding Principles for Timely Administration of Medications](#)

## 7. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated by AMS committee reporting the NAPS audit, NAUPS program and SAPS survey data.

## 8. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.15 Antimicrobial stewardship
- The health service organisation has an antimicrobial stewardship program that:
  - a. Includes an antimicrobial stewardship policy
  - b. Provides access to, and promotes the use of, current evidence-based Australian therapeutic guidelines and resources on antimicrobial prescribing
  - c. Has an antimicrobial formulary that includes restriction rules and approval processes
  - d. Incorporates core elements, recommendations and principles from the current Antimicrobial Stewardship Clinical Care Standard<sup>117</sup>
- Standard 3 – Action 3.16
- The antimicrobial stewardship program will:
  - a. Review antimicrobial prescribing and use
  - b. Use surveillance data on antimicrobial resistance and use to support appropriate prescribing
  - c. Evaluate performance of the program, identify areas for improvement, and take action to improve the appropriateness of antimicrobial prescribing and use
  - d. Report to clinicians and the governing body regarding
    - compliance with the antimicrobial stewardship policy
    - antimicrobial use and resistance
    - appropriateness of prescribing and compliance with current evidence-based Australian therapeutic guidelines or resources on antimicrobial prescribing
- Standard 4 – Action 4.15 High-risk medicines
- The health service organisation:
  - a. Identifies high-risk medicines used within the organisation
  - b. Has a system to store, prescribe, dispense and administer high-risk medicines safely

## 9. References

1. eTG Therapeutic Guidelines - [Antibiotic](#)

## 10. Document owner

### Director Clinical Services

Enquiries relating to this guideline may be directed to:

Title: Deputy Director Clinical Services  
 Department: Hospital Wide  
 Email: [xin.chua@health.wa.gov.au](mailto:xin.chua@health.wa.gov.au)

## 11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

| Version | Effective from | Effective to | Amendment(s)     |
|---------|----------------|--------------|------------------|
| V2      | 07/11/2023     | 07/11/2026   | Scheduled review |

## 12. Authorisation

This guideline has been authorised and issued by the Director Clinical Services.

|                    |                                             |
|--------------------|---------------------------------------------|
| Authorised by      | Dr Alison Parr – Director Clinical Services |
| Authorisation date | 07/11/2023                                  |
| Published date     | 07/11/2023                                  |

## Appendix 1: Abdominal surgery

### Surgical prophylaxis **not** indicated:

- endoscopic gastroduodenal and oesophageal procedures rhytidectomy

### Surgical prophylaxis indicated:

- non-endoscopic gastroduodenal or oesophageal procedures that **enter** the gastrointestinal tract lumen
- non-endoscopic gastroduodenal or oesophageal procedures that **do not enter** the gastrointestinal tract lumen (e.g. anti-reflux procedures), **ONLY** if the patient has risk factors:
  - morbid obesity
  - gastric outlet obstruction
  - reduced gastric acidity or motility
  - gastrointestinal bleeding
  - malignancy or perforation

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

PLUS

vancomycin 15 mg/kg IV, started within the 15-120 minutes before surgical incision (recommended rate 10 mg/minute)

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 2: Appendicectomy procedures

**Surgical prophylaxis indicated:**

- all appendicectomy procedures

**Surgical prophylaxis may not be required:**

- patients receiving antibiotic therapy for acute intra-abdominal infection. Adjust timing of antibiotics prior to surgery.

If patient is not already receiving antibiotic therapy for appendicitis (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
PLUS  
metronidazole 500mg IV, within the 120 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

metronidazole 500 mg IV, within the 120 minutes before surgical incision  
PLUS  
gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed for patients with uncomplicated appendicitis**

**For patients with complicated appendicitis, consider ceasing therapy 5 days after adequate surgical source control**

## Appendix 3: Breast surgery

### Surgical prophylaxis not indicated:

- diagnostic excisional biopsy
- stand-alone sentinel node biopsy
- lumpectomy (with or without needle or wire localisation)

For prophylaxis for spinal surgery (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to cefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use vancomycin **instead** of cefazolin.

### Surgical prophylaxis indicated:

- reduction mammoplasty
- simple mastectomy
- wide local excision
- axillary lymph node clearance
- nipple surgery
- all repeat or revision procedures
- prosthetic breast reconstruction surgery (prosthetic implant or acellular dermal matrix)
- autologous breast reconstruction surgery
- breast augmentation surgery

**Do NOT use topical antimicrobials or antiseptics to irrigate the surgical pocket at the time of breast implantation**

**Antibiotic prophylaxis may continue for a maximum of 24 hours (2 doses post-operatively). Do not exceed 24 hours of antibiotic prophylaxis.**

## Appendix 4: Biliary surgery, including laparoscopic surgery

### Surgical prophylaxis not indicated:

- laparoscopic surgery in the absence of risk factors for postoperative infection (see below)

### Surgical prophylaxis indicated:

- laparoscopic surgery **ONLY** if patient has risk factors for postoperative infection
  - age >70 years
  - diabetes
  - obstructive jaundice
  - common bile duct stones
  - acute cholecystitis
  - non-functioning gallbladder
- Open cholecystectomy

### Surgical prophylaxis may not be required:

- patients receiving antibiotic therapy for acute intra-abdominal infection. Adjust timing of antibiotics prior to surgery.

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use vancomycin **instead** of cefazolin.

gentamicin 2 mg/kg intravenously over 3 to 5 minutes, within the 120 minutes before surgical incision

PLUS

vancomycin 15 mg/kg IV, started within the 15-120 minutes before surgical incision (recommended rate 10 mg/minute)

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**



**Appendix 5: Colorectal surgery****Surgical prophylaxis indicated:**

- non-endoscopic colorectal procedures

**Surgical prophylaxis may not be required:**

- patients receiving antibiotic therapy for acute intra-abdominal infection. Adjust timing of antibiotics prior to surgery.

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
PLUS  
metronidazole 500mg IV, within the 120 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

metronidazole 500 mg IV, within the 120 minutes before surgical incision  
PLUS  
gentamicin 2 mg/kg intravenously over 3 to 5 minutes, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 6: Ear, nose and throat surgery

### Surgical prophylaxis **not** indicated:

- nasal packing or a tamponade device *in situ* following epistaxis
- uncomplicated nose or sinus surgery (including endoscopic procedures)
- uncomplicated ear surgery
- otoplasty
- stapedectomy
- tonsillectomy [see specific patient considerations]
- adenoidectomy [see specific patient considerations]

### Surgical prophylaxis indicated:

- major ear surgery
- complex septorhinoplasty
- revision sinus surgery
- laryngectomy (primary or salvage)
- tympanomastoid surgery
- hearing implant procedures, including cochlear implant procedures

### Specific patient considerations:

- patients undergoing tonsillectomy or adenoidectomy with specific cardiac conditions (refer to “*prevention of infective endocarditis*” in Therapeutic Guidelines)

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
PLUS  
metronidazole 500mg IV, within the 120 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to above regimen:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

clindamycin 600mg IV, within the 120 minutes before surgical incision  
**PLUS** for laryngectomy or tympanomastoid surgery  
gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed unless EXCEPT in patients undergoing laryngectomy**

**Postoperative antibiotic doses can be considered for up to a maximum of 24 hours for patients undergoing laryngectomy**

**Appendix 7: Endoscopic retrograde cholangiopancreatography:****Surgical prophylaxis not indicated:**

- ERCP with evidence of biliary tract obstruction if complete biliary drainage is likely to be achieved
- other ERCP procedures if the patient does **not** have communicating pancreatic cysts or pseudocysts

**Surgical prophylaxis indicated:**

- ERCP involving transpapillary or transmural drainage of pseudocyst
- ERCP with evidence of biliary tract obstruction **ONLY** IF complete biliary drainage may not be achieved
- other ERCP procedures **ONLY** if the patient has communicating pancreatic cysts or pseudocysts

If prophylaxis is indicated, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 8: Endoscopic ultrasound

**Surgical prophylaxis not indicated:**

- diagnostic EUS
- EUS-FNA of solid lesions

**Surgical prophylaxis indicated:**

- EUS-FNA of cystic lesions

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**), use:

metronidazole 500 mg IV, within the 120 minutes before surgical incision  
**PLUS**  
cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**) use:

metronidazole 500 mg IV, within the 120 minutes before surgical incision  
**PLUS**  
gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 9: Endoscopic urological procedures

If prophylaxis is indicated for endoscopic urological procedures, in the absence of culture and susceptibility test results, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

If gentamicin is contraindicated, use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed.**

## Appendix 10: Gastrostomy or jejunostomy tube insertion

**Surgical prophylaxis indicated:**

- PEG or PEJ tube insertion
- PRG or PRJ tube insertion

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to cefazolin:

vancomycin 15 mg/kg intravenously, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use vancomycin **instead** of cefazolin.

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 11: Gynaecological surgery

Investigate patients for sexually transmitted infections (STIs):

- if they have symptoms of an STI
- before insertion of an intrauterine contraceptive device
- before a transcervical procedure, including surgical termination of pregnancy and hysteroscopy.

If the results of investigations are positive, provide appropriate treatment before the procedure.

### Surgical prophylaxis not indicated:

- laparoscopic procedures that do not enter the bowel or vagina
- hysteroscopy, operative or diagnostic
- dilation and curettage, with the exception of surgical termination of pregnancy
- endometrial biopsy or ablation
- insertion of an intrauterine device
- cervical tissue excision procedure (eg LLETZ, biopsy, endocervical curettage)
- autologous mid-urethral sling procedures

### Surgical prophylaxis indicated:

- Hysterectomy
- gynaecological–oncological procedures
- gynaecological laparotomy procedures
- synthetic mid-urethral sling procedures
- pelvic organ prolapse procedures
- surgical termination of pregnancy if not investigated for STIs before the procedure

### Specific patient considerations:

- Patients with specific cardiac conditions may require additional antibiotics for prophylaxis against enterococcal endocarditis (refer to “*endocarditis prophylaxis for gastrourinary and gastrointestinal tract procedures*” in Therapeutic Guidelines)

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity** use:

gentamicin 2 mg/kg intravenously over 3 to 5 minutes, within the 120 minutes before surgical incision

**PLUS**

clindamycin 600 mg IV, within the 120 minutes before surgical incision;

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 12: Head and neck surgery

**Surgical prophylaxis not indicated:**

- thyroidectomy
- simple lymph node excision (including submandibular lymph node excision)
- parotidectomy
- clean procedures not listed below

**Surgical prophylaxis indicated:**

- procedures involving insertion of prosthetic material
- extensive neck dissection for malignancy
- debulking or reconstructive surgery for malignancy
- clean-contaminated procedure not listed above

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**), use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

**PLUS** for incisions through mucosal surfaces  
metronidazole 500 mg IV, within the 120 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

clindamycin 600 mg IV, within the 120 minutes before surgical incision

**PLUS** for extensive neck dissection, or debulking or reconstructive surgery

gentamicin 2 mg/kg intravenously over 3 to 5 minutes, within the 120 minutes before surgical incision

**Antibiotic prophylaxis (IV or oral) should not continue beyond 24 hours**



## Appendix 13: Hernia repair

### Surgical prophylaxis indicated:

- hernia repair with or without prosthetic material (mesh)

If **entry into the bowel lumen is not expected** (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

If **entry into the bowel lumen is not expected** for patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to cefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

If **entry into the bowel lumen is not expected**, for patients with immediate **severe** or immediate **severe penicillin hypersensitivity** use vancomycin as monotherapy (see dosage above).

If **entry into the bowel lumen is expected** (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
PLUS  
metronidazole 500mg IV, within the 120 minutes before surgical incision

If **entry into the bowel lumen is expected** for patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to cefazolin:

vancomycin (adult and child) 15 mg/kg intravenously, started within the 120 minutes before surgical incision (recommended rate 10 mg/minute)

If **entry into the bowel lumen is expected** for patients with **immediate severe** or immediate **severe penicillin hypersensitivity** use:

metronidazole 500 mg IV, within the 120 minutes before surgical incision  
PLUS  
gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add vancomycin to metronidazole and gentamicin.

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 14: Implantable cardiac device insertion

**Surgical prophylaxis indicated:**

- insertion of an implantable cardiac device (eg permanent pacemaker device, cardioverter defibrillator, cardiac resynchronisation device, ventricular assist device)

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to cefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

**PLUS**

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed.**

## Appendix 15: Open or laparoscopic urological procedures

### Entry into the bowel lumen NOT expected

If prophylaxis is indicated for an open or laparoscopic urological procedure in which entry into the bowel lumen is not expected (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**), use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
**PLUS**  
 gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before the procedure

If gentamicin is contraindicated seek expert advice.

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before the procedure  
**PLUS EITHER**  
 vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)  
**OR**  
 clindamycin 600mg IV, started within the 120 minutes before the procedure

In cases of inadvertent rectal injury, administer metronidazole immediately. Use:

metronidazole 500mg IV, as a single dose

### Entry into the bowel lumen IS expected (e.g. ileal conduit, rectocele repair):

If prophylaxis is indicated for an open or laparoscopic urological procedure in which entry into the bowel lumen is expected (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**), use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
**PLUS**  
 metronidazole 500mg IV, within the 120 minutes before surgical incision

If gentamicin is contraindicated seek expert advice.

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before the procedure  
**PLUS**  
 metronidazole 500mg IV, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed.**

## Appendix 16: Neurosurgery procedures

### Surgical prophylaxis **not** indicated:

- cerebrospinal fluid leakage following trauma
  - vaccinate against *Streptococcus pneumoniae* to protect against the development of pneumococcal meningitis. See the *Australian Immunisation Handbook*

### Surgical prophylaxis indicated:

- intracranial shunt insertion
- pressure monitor insertion
- craniotomy
- microsurgery
- procedures involving insertion of prosthetic material
- re-exploration procedures
- external ventricular drain insertion

For prophylaxis for spinal surgery (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** **add** to cefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use vancomycin **instead** of cefazolin.

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed.**

**Ventricular drains that remain *in situ* do not justify extending the duration of antibiotic prophylaxis postoperatively.**

## Appendix 17: Obstetric surgery

### Surgical prophylaxis indicated:

- caesarean section, elective or nonelective (even if patient is receiving intrapartum prophylaxis against *Streptococcus agalactiae* or treatment for intra-amniotic infection).
- Assisted instrumental delivery (see below)

### Assisted instrumental delivery

SINGLE dose of 1.2 g amoxicillin/clavulanic acid IV, as soon as possible  
AFTER birth and no more than 6 hours after birth

### Other procedures

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add tocefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before  
surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, who are neither colonised nor infected with **MRSA**, nor at increased risk of being colonised or infected with **MRSA** use:

gentamicin 2 mg/kg intravenously over 3 to 5 minutes, within the 120  
minutes before surgical incision  
**PLUS**  
clindamycin 600 mg IV, within the 120 minutes before surgical incision;

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, who are colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** use:

gentamicin 2 mg/kg intravenously over 3 to 5 minutes, within the 120  
minutes before surgical incision  
**PLUS**  
vancomycin 15 mg/kg intravenously, started within the 15 to 120 minutes  
before surgical incision (recommended rate 10 mg/minute)

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 18: Ophthalmic surgery

The aim of antibiotic prophylaxis is to reduce acute postoperative endophthalmitis.

Prevention of endophthalmitis involves a combination of preoperative screening, antisepsis, intraoperative techniques, and postoperative patient care.

Active conjunctivitis, dacryocystitis or blepharitis must be treated and resolved before surgery.

Antibiotics (particularly vancomycin) in irrigation solutions should not be used.

Intracameral administration (injection into the anterior chamber of the eye) of antibiotics at the end of surgery is the most effective method of reducing the risk of endophthalmitis after cataract surgery. Use:

cefazolin 1 mg intracamerally, as a single dose at the end of surgery

**Providing postoperative topical antibiotics lack adequate evidence.**

If postoperative topical antibiotics are considered necessary, chloramphenicol is recommended. Use:

chloramphenicol 0.5% eye drops, 1 drop into the operated eye, four times a day for a maximum of 7 days.

Tobramycin eye drops should only be considered for patients hypersensitive to chloramphenicol but this does not have activity against Gram-positive organisms.

**There is a lack of evidence to support use of topical quinolones, seek expert advice.**

## Appendix 19: Oral maxillofacial procedures

### Surgical prophylaxis not indicated:

- clean or clean-contaminated procedures
- procedures involving insertion of dental implants

### Surgical prophylaxis indicated:

- procedures involving insertion of prosthetic material, with the exception of dental implants
- open reduction and internal fixation of mandibular fractures or midfacial (eg Le Fort or zygomatic) fractures
- intraoral bone grafting procedures
- orthognathic surgery (major jaw realignment surgery)
- cleft lip and palate repairs

### Specific patient considerations:

- patients a procedure that involves manipulation of the gingival or periapical tissue or perforation of the oral mucosa, prophylaxis against streptococcal endocarditis is required (refer to “*endocarditis prophylaxis in dental procedures*” in therapeutic guidelines)

For prophylaxis for procedures involving incision through the **oral mucosa only** (e.g. cleft lip and palate repairs), use:

benzylpenicillin 1.2g IV, within the 60 minutes before surgical prophylaxis

For prophylaxis for procedures involving incision through the **skin and oral mucosa** (e.g. temporomandibular joint replacement), use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

PLUS

metronidazole 500mg IV. within the 120 minutes before surgical incision

For patients with **immediate non-severe** or **delayed non-severe penicillin hypersensitivity**, use cefazolin plus metronidazole (as above).

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

clindamycin 600mg IV, within the 120 minutes before surgical incision

**Postoperative antibiotic doses can be considered for up to a maximum of 24 hours. Do not exceed prophylaxis (IV or oral) beyond 24 hours.**

## Appendix 20: Orthopaedic surgery

### Surgical prophylaxis **not** indicated:

- routine arthroscopy procedures not involving insertion of prosthetic material or avascular tissue

### Surgical prophylaxis indicated:

- prosthetic large joint replacement
- procedures involving insertion of prosthetic or allograft material
- **internal fixation of fractures of large bones**
- open fractures

For prophylaxis for orthopaedic surgery (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** (e.g. patients undergoing a joint arthroplasty procedure that is a reoperation [return to theatre or early revision]; **add** to cefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use vancomycin **instead** of cefazolin.

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed EXCEPT in total knee arthroplasty.**

**For total knee arthroplasty, postoperative antibiotic doses can be considered for up to a maximum of 24 hours. Do not exceed prophylaxis (IV**

For open fractures, refer to “open fractures” in Therapeutic Guidelines: antibiotics



## Appendix 21: Skin and soft tissue procedures

**Surgical prophylaxis not indicated:**

- blepharoplasty
- rhytidectomy
- other clean or clean–contaminated procedures of skin and subcutaneous tissue, including those that breach the oral mucosa

**Surgical prophylaxis indicated:**

- ablative laser facial resurfacing procedures, particularly for patients who have had previous orofacial HSV infection: consider perioperative antiviral therapy
- traumatic wounds requiring surgical management
- bites and clenched-fist injury infections requiring surgical management

**Appendix 22: Small intestinal surgery****Surgical prophylaxis not indicated:**

- endoscopic small intestine procedures

**Surgical prophylaxis indicated:**

- non-endoscopic small intestine procedures

**Surgical prophylaxis may not be required:**

- patients receiving antibiotic therapy for acute intra-abdominal infection. Adjust timing of antibiotics prior to surgery.

If prophylaxis is indicated and the small intestine is **not obstructed** (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

If prophylaxis is indicated and the small intestine is **obstructed** (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision  
PLUS  
metronidazole 500mg IV, within the 120 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, whether or not the small intestine is obstructed, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision  
PLUS  
metronidazole 500 mg IV, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 23: Spinal surgery

**Surgical prophylaxis indicated:**

- all spinal surgery procedures

For prophylaxis for spinal surgery (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA** add to cefazolin:

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use vancomycin **instead** of cefazolin.

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed**

## Appendix 24: Transperineal prostate biopsy

For prophylaxis for transperineal prostate biopsy, in the absence of culture and susceptibility test results (**including patients with immediate non-severe or delayed non-severe** penicillin hypersensitivity), use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before the procedure

**PLUS EITHER**

vancomycin 15 mg/kg IV, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

**OR**

clindamycin 600mg IV, started within the 120 minutes before the procedure

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed.**

### Transrectal prostate biopsy

Asymptomatic bacteriuria and bacteraemia are common after transrectal prostatic biopsy; less commonly, symptomatic postoperative urinary tract infection, prostatitis, sepsis and, rarely, septic shock occur.

For patients treated for bacteriuria preoperatively, modify the choice of surgical antibiotic prophylaxis based on the results of culture and susceptibility testing.

For prophylaxis for transrectal prostate biopsy, in the absence of culture and susceptibility test results, use:

ciprofloxacin 500 mg orally, as a single dose, 120 minutes before the procedure.

Seek expert advice for alternative prophylactic antibiotic regimes in patients at high risk of carriage of ciprofloxacin resistance isolates.

**Appendix 25: Transurethral resection of the prostate**

For patients treated for bacteriuria preoperatively, modify the choice of surgical antibiotic prophylaxis based on the results of culture and susceptibility testing.

For prophylaxis for transurethral resection of the prostate, in the absence of culture and susceptibility test results, use:

gentamicin 2 mg/kg\* IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

If gentamicin is contraindicated, use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed.**

## Appendix 26: Urological surgery

### Preoperative bacteriuria

#### Screening for perioperative bacteriuria

##### Uncomplicated cystoscopic diagnostic procedures:

- Do not screen for bacteriuria because the risk of postoperative infection is low. However, if the results of urinalysis suggest urinary tract infection, urine culture should be performed.

##### Other elective urological procedures that enter the urinary tract:

- Perform preoperative urine culture. In catheterised patients, collect samples using a new catheter to avoid contamination of the sample by organisms colonising the old catheter.

If preoperative screening is not possible before an **immediate operation**, empirical treatment may be required.

##### Treating for perioperative bacteriuria confirmed by screening

Use the results of culture and susceptibility testing to choose the most appropriate regimen (**see eTG: Antibiotic guidelines, Acute cystitis in adults and Acute cystitis in children**).

Alternatively, treat with a single preoperative dose of gentamicin (as below).

For patients with chronic infection of the urinary tract, results of culture and susceptibility testing should guide antibiotic choice—seek expert advice.

##### General principles

Preoperative treatment of bacteriuria does not negate the need for surgical antibiotic prophylaxis. An exception is when gentamicin is given preoperatively to treat bacteriuria and is also indicated as prophylaxis

Patients with specific cardiac conditions (**see eTG, Antibiotic Guidelines, Endocarditis prophylaxis for genitourinary and gastrointestinal tract procedures**).

**Pre-operative considerations:** urine cultures before elective urological procedures involving manipulation of the urinary tract. Treat significant bacteriuria even if asymptomatic. Use culture and sensitivity results for treatment regime (**see eTG: Antibiotic guidelines, Acute cystitis in adults and Acute cystitis in children**).

**Postoperative considerations:** Do not continue antibiotic prophylaxis until the urinary catheter is removed – unless it offers a benefit for some high-risk patients.

If immediate operation is required with clinical evidence of a UTI or confirmed bacteriuria, treat with a single dose of gentamicin. Use:

gentamicin 3mg/kg\*, as a single preoperative dose

*\*a higher dose is needed in patients with systemic symptoms (e.g. severe pyelonephritis)*

#### **Surgical prophylaxis not indicated:**

- urodynamic studies
- extracorporeal shock-wave lithotripsy
- open or laparoscopic urological procedures in which the urinary tract is not entered (eg vasectomy, scrotal surgery, varicocele ligation) **and** prosthetic material is not implanted

#### **Surgical prophylaxis indicated:**

- prostate fiducial marker insertion
- endoscopic intrarenal and ureteric stone procedures (e.g. percutaneous nephrolithotomy, pyeloscopy for ureteric or kidney stones)
- ureteroscopy procedures
- Other endoscopic or uncomplicated cystoscopic diagnostic procedures **ONLY** If there are risks for postoperative infection (e.g. urinary tract obstruction or abnormalities, urinary stones, indwelling or externalised catheters)
- transurethral resection of the prostate
- transrectal prostate biopsy
- transperineal prostate biopsy
- open or laparoscopic urological procedures involving implantation of prosthetic material (eg penile prostheses, artificial urinary sphincters, mesh)
- open or laparoscopic urological procedures where the urinary tract is entered

#### **Individual patient-based decision**

- specific risks for postoperative infection (e.g. lithotripsy in patients with an internal stent, nephrostomy tube or indwelling catheter in situ)
- immediate operation and bacteriuria cannot be excluded

## Appendix 27: Vascular surgery

**Surgical prophylaxis not indicated:**

- varicose veins procedures
- brachial or carotid artery procedures, **unless** prosthetic material is inserted

**Surgical prophylaxis indicated:**

- limb amputation
- vascular reconstructive surgery involving abdominal aorta or lower limbs and limb amputation surgery

**Surgical prophylaxis may not be required:**

- patient already receiving antibiotic treatment for an established infection with activity against the organism(s) most likely to cause postoperative infection (this should include anaerobic cover for ischaemic limb amputation).
  - Adjust timing of antibiotics prior to surgery
  - Intra-operative re-dosing may be required

If prophylaxis is indicated (including patients with immediate **non-severe** or delayed **non-severe penicillin hypersensitivity**) use:

cefazolin 2 g IV, within the 60 minutes before surgical incision

For amputation of an ischaemic limb **ADD**

metronidazole 500mg IV, within the 120 minutes before surgical incision

For patients colonised or infected with methicillin-resistant *Staphylococcus aureus* (**MRSA**), or at increased risk of being colonised or infected with **MRSA add** to the above regimen:

vancomycin 15 mg/kg intravenously, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

For patients with immediate **severe** or delayed **severe penicillin hypersensitivity**, use:

vancomycin 15 mg/kg intravenously, started within the 15 to 120 minutes before surgical incision (recommended rate 10 mg/minute)

**PLUS**

gentamicin 2 mg/kg IV, over 3 to 5 minutes, within the 120 minutes before surgical incision

**PLUS for amputation of ischaemic limb**

metronidazole 500 mg IV, within the 120 minutes before surgical incision

**Do NOT continue antibiotic prophylaxis (IV or oral) once procedure is completed for patients with uncomplicated appendicitis**